

1. Perturbed fixed point $\Leftrightarrow$

Suppose $G : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a contractive map with Lipschitz constant (for some norm) of $\alpha < 1$. Therefore it has a unique fixed point x^* . Let $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be Lipschitz with constant β , and consider the perturbed function $\hat{G}_\epsilon : \mathbb{R}^n \rightarrow \mathbb{R}^n$. Show that for small enough ϵ , $\hat{G}_\epsilon$ also has a unique fixed point $\hat{x}^*$, and give a bound on the distance between x^* and $\hat{x}^*$.

We know that $\hat{G}_\epsilon(x) = G(x) + \epsilon F(x)$

Our first action is to prove that $\hat{G}_\epsilon(x)$ is a contraction (for small enough ϵ), which will prove that it has a unique fixed point ($\hat{x}^*$)

For any $x, y \in \mathbb{R}^n$, using the Triangle inequality + the Lipschitz-ness of G and F

$$\begin{aligned}\|\hat{G}_\epsilon(x) - \hat{G}_\epsilon(y)\| &= \|G(x) - G(y) + \epsilon(F(x) - F(y))\| \\ &\leq \|G(x) - G(y)\| + |\epsilon| \|F(x) - F(y)\| \leq (\alpha + |\epsilon|\beta) \|x - y\|\end{aligned}$$

Therefore, the Lipschitz Constant for $\hat{G}_\epsilon(x)$, which we will call $\hat{\alpha} = \alpha + |\epsilon|\beta$, we need $\hat{\alpha} < 1$, which is only possible when

$$|\epsilon| < \frac{1 - \alpha}{\beta}$$

For any such ϵ that satisfies the above, $\hat{G}_\epsilon$ is a contraction on $\mathbb{R}^n$ with constant $\hat{\alpha} < 1$, so by the Banach fixed-point theorem it has a unique fixed point $\hat{x}^*$.

To give the bound on the distance (which is $\|x^* - \hat{x}^*\|$), notice that by definition of the "fixed point":

$$x^* = G(x^*), \hat{x}^* = \hat{G}_\epsilon(x) = G(\hat{x}^*) + \epsilon F(\hat{x}^*)$$

So, doing the subtraction, we get that

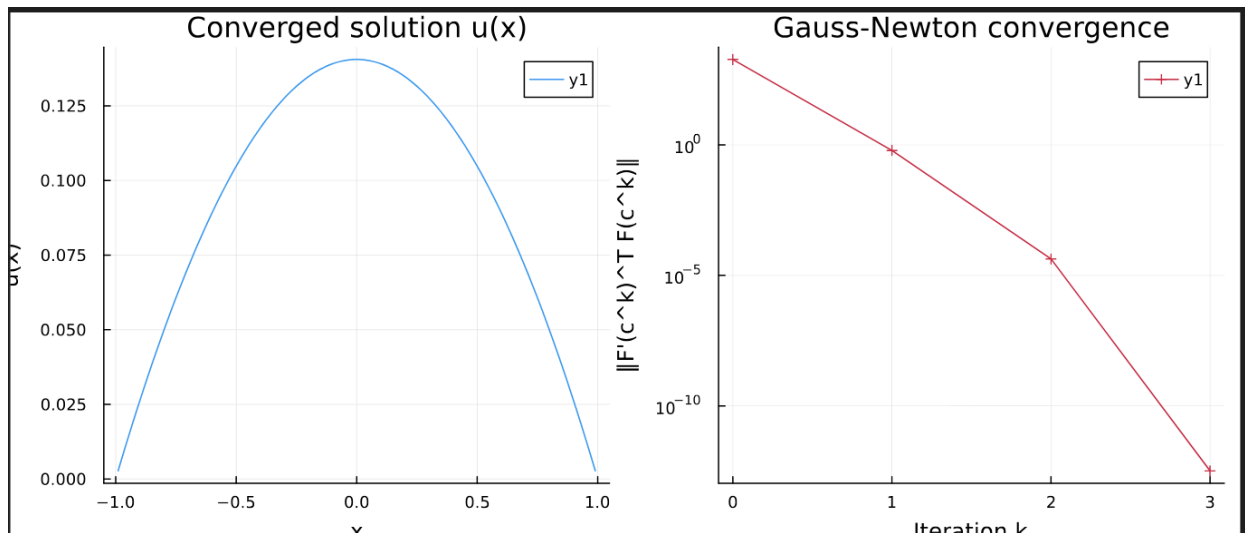
$$x^* - \hat{x}^* = G(x^*) - G(\hat{x}^*) - \epsilon F(\hat{x}^*) = G(x^*) - G(\hat{x}^*) - \epsilon F(x^*) + \epsilon(F(x^*) - F(\hat{x}^*))$$

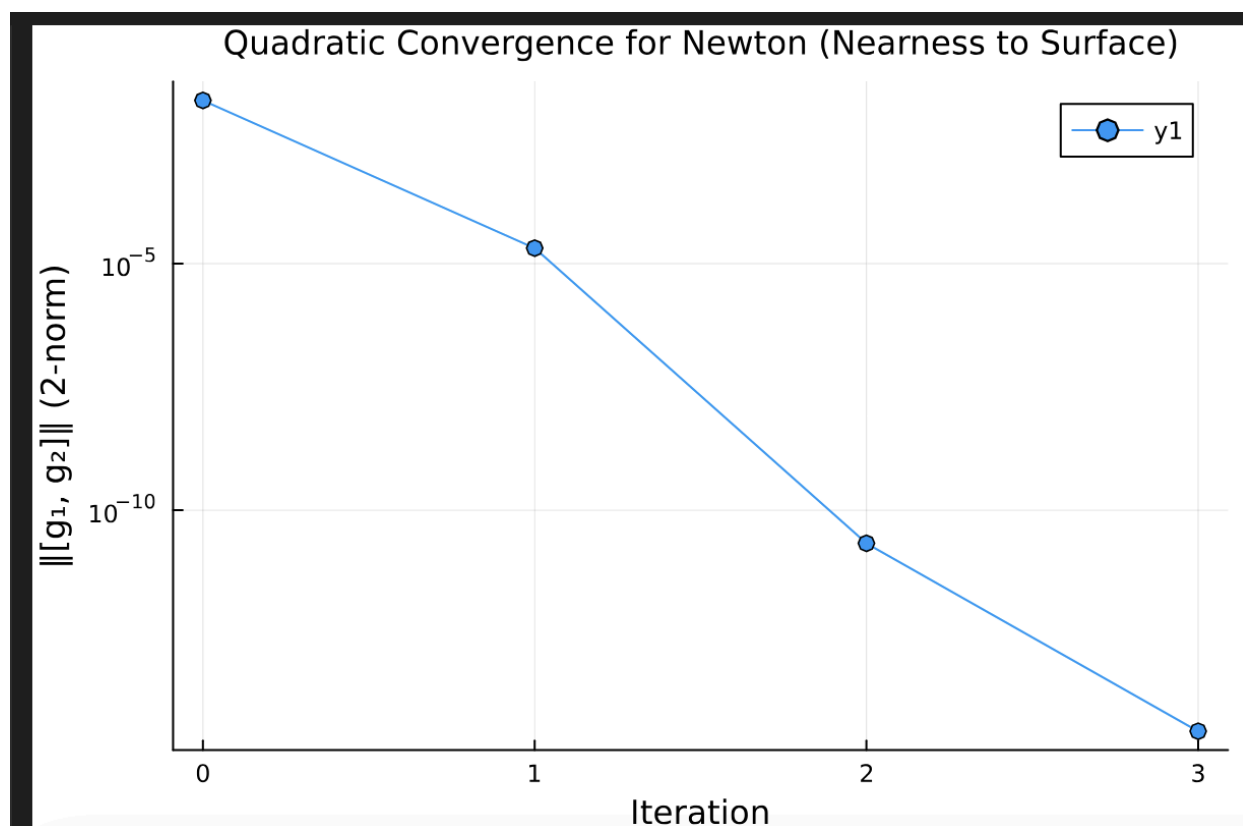
By taking norms, using norm properties & triangle inequality, and the Lipschitz Properties for F where it has Lipschitz Constant β and for G where it has Lipschitz Constant α :

$$\begin{aligned} \|x^* - \hat{x}^*\| &\leq \|G(x^*) - G(\hat{x}^*)\| + |\epsilon| \|F(x^*)\| + |\epsilon| \|F(x^*) - F(\hat{x}^*)\| \\ &\leq \alpha \|x^* - \hat{x}^*\| + |\epsilon| \|F(x^*)\| + |\epsilon|\beta \|x^* - \hat{x}^*\| = (\alpha + |\epsilon|\beta) \|x^* - \hat{x}^*\| + |\epsilon| \|F(x^*)\| \end{aligned}$$

We also know that When we rearrange, we get the final answer:

$$\|x^* - \hat{x}^*\| \leq \frac{|\epsilon| \|F(x^*)\|}{1 - \alpha - |\epsilon|\beta}$$





```
> Closest point: (2.999491034956831, 1.9993213096792508, 0.0001768938306988543  
4)  
Distance: 0.0008665779791720156
```